



6G5Z0050 : Maths for Data Science

DURATION : 2 hours

Instructions to candidates:

Answer all the questions.

You must show all working to gain full marks.

Permitted materials:

Calculators are permitted. You may bring one double-sided A4 sheet of notes into the exam with you. A standard formula sheet will also be provided.

DO NOT REMOVE THIS PAPER FROM THE EXAMINATION VENUE

1. (a) The times, in seconds, taken by 20 people to solve a simple puzzle were

17 19 22 26 28 31 34 36 38 39 41 42 43 47 50 51 53 55 57 58

- (i) Calculate the mean and standard deviation of these times. (6 marks)

Later, after eating a protein bar, the same group of people were given a different puzzle to solve, and the mean solve time was 25.1 seconds, with a standard deviation of 4.5 seconds.

- (ii) Comment on the meaning of these results. (3 marks)

- (iii) Give two reasons why these results may not be useful in determining the effect of the protein bar on puzzle solving ability. (4 marks)

- (b) As part of a larger study, the same experiment was performed on a larger cohort, and the results were found to be normally distributed. Let $X \sim N(41, 10.5)$ represent the performance of 100 people before the protein bar. After eating the bar, the mean solve time was 38.5 seconds.

Perform a statistical test at the 5% significance level to determine whether their performance has improved significantly. State your hypotheses and perform the test. (12 marks)

Total marks for question 1: [25]

2. (a) Find the determinant of the matrix $M = \begin{pmatrix} -1 & 2 & 0 \\ 7 & 4 & 3 \\ 5 & 2 & -4 \end{pmatrix}$ (5 marks)

- (b) (i) Calculate the dot product of the two vectors $\mathbf{u} = (2, 3)$ and $\mathbf{w} = (1, 6)$. (2 marks)

- (ii) Calculate the magnitude of the vector $\mathbf{u} + \mathbf{v}$. (4 marks)

- (c) The following data was collected about the ages, and scores from the first three darts thrown, of participants in a friendly darts match.

Age (X)	Score (Y)
26	60
29	68
30	58
35	40

Calculate the population covariance matrix for this data. (12 marks)

Total marks for question 2: [23]

3. (a) A large office building buys pens from two suppliers - Office World, and Stapler King. The office manager knows that 10% of the pens supplied by Office World and 5% of those supplied by Stapler King are defective.

100 pens are stored together in a box, of which 65 are from Office World and the remainder from Stapler King. A member of staff selects a pen at random from the box.

- (i) Find the probability that the pen is from Office World. (2 marks)
- (ii) How many of the pens in the box would you expect to be both from Stapler King and defective? (3 marks)
- (iii) Find the probability that a randomly chosen pen from the box is defective. (4 marks)

A member of staff who took a pen from the box found that their pen was defective, and complained to the college management about the quality of pens from Office World.

- (iv) Given that the pen was defective, find the probability that it was supplied by Office World. (5 marks)

- (b) The office manager wishes to do some more accurate modelling of the pen situation, and suggests a binomial distribution could be used to create a model for X , the number of defective pens they might expect to have out of the 10,000 pens they have ordered from Office World. Is this a suitable model, and if so how would you express it? Explain your answer. (6 marks)

Total marks for question 3: [20]

4. (a) It is believed that a high school student's favourite film genre is related to their favourite school subject. Data scientists wish to test this theory by surveying 500 school students from the region's 10 schools, and asking them to indicate their favourite film genre and favourite subject from a selection of options.
- (i) The data scientists need to decide between collecting a simple random sample or a cluster sample. Describe what each involves, briefly indicate how you might implement it, and compare the two methods. (6 marks)

(b) The survey was conducted, and the results are indicated in the table below.

	Comedy	Action	Romance	Thriller
Maths	51	52	37	55
Sports	59	63	41	33
Geography	35	31	28	15

The data science team wish to determine whether there is any association between favourite subject and favourite film genre by using a chi-squared test for independence at the 5% significance level.

- (i) State the null and alternative hypotheses for this test. (2 marks)
- (ii) Write down the number of degrees of freedom for this table. (2 marks)
- (iii) Calculate the chi-squared test statistic for this data. (15 marks)
- (iv) Write down the conclusion to the test and explain what it means. (3 marks)

Total marks for question 4: [33]

1 Reference - Statistical Tables

1.1 Chi-Squared Test Probability Table

DF	1.00	0.99	0.98	0.95	0.90	0.75	0.50	0.25	0.10	0.05	0.03	0.01	0.01
1	0.0000	0.0002	0.0010	0.0039	0.0158	0.1015	0.4549	1.3233	2.7055	3.8415	5.0239	6.6349	7.8794
2	0.0100	0.0201	0.0506	0.1026	0.2107	0.5754	1.3863	2.7726	4.6052	5.9915	7.3778	9.2103	10.597
3	0.0717	0.1148	0.2158	0.3519	0.5844	1.2125	2.3660	4.1083	6.2514	7.8147	9.3484	11.345	12.838
4	0.2070	0.2971	0.4844	0.7107	1.0636	1.9226	3.3567	5.3853	7.7794	9.4877	11.143	13.277	14.860
5	0.4117	0.5543	0.8312	1.1455	1.6103	2.6746	4.3515	6.6257	9.2364	11.071	12.833	15.086	16.750
6	0.6757	0.8721	1.2373	1.6354	2.2041	3.4546	5.3481	7.8408	10.645	12.592	14.449	16.812	18.548
7	0.9893	1.2390	1.6899	2.1674	2.8331	4.2549	6.3458	9.0372	12.017	14.067	16.013	18.475	20.278
8	1.3444	1.6465	2.1797	2.7326	3.4895	5.0706	7.3441	10.219	13.362	15.507	17.535	20.090	21.955
9	1.7349	2.0879	2.7004	3.3251	4.1682	5.8988	8.3428	11.389	14.684	16.919	19.023	21.666	23.589
10	2.1559	2.5582	3.2470	3.9403	4.8652	6.7372	9.3418	12.549	15.987	18.307	20.483	23.209	25.188
11	2.6032	3.0535	3.8158	4.5748	5.5778	7.5841	10.341	13.701	17.275	19.675	21.920	24.725	26.757
12	3.0738	3.5706	4.4038	5.2260	6.3038	8.4384	11.340	14.845	18.549	21.026	23.337	26.217	28.300
13	3.5650	4.1069	5.0088	5.8919	7.0415	9.2991	12.340	15.984	19.812	22.362	24.736	27.688	29.819
14	4.0747	4.6604	5.6287	6.5706	7.7895	10.165	13.339	17.117	21.064	23.685	26.119	29.141	31.319
15	4.6009	5.2294	6.2621	7.2609	8.5468	11.037	14.339	18.245	22.307	24.996	27.488	30.578	32.801
16	5.1422	5.8122	6.9077	7.9617	9.3122	11.912	15.339	19.369	23.542	26.296	28.845	32.000	34.267
17	5.6972	6.4078	7.5642	8.6718	10.085	12.792	16.338	20.489	24.769	27.587	30.191	33.409	35.718
18	6.2648	7.0149	8.2308	9.3905	10.865	13.675	17.338	21.605	25.989	28.869	31.526	34.805	37.156
19	6.8440	7.6327	8.9065	10.117	11.651	14.562	18.338	22.718	27.204	30.144	32.852	36.191	38.582
20	7.4338	8.2604	9.5908	10.851	12.443	15.452	19.337	23.828	28.412	31.410	34.170	37.566	39.997
21	8.0337	8.8972	10.283	11.591	13.240	16.344	20.337	24.935	29.615	32.671	35.479	38.932	41.401
22	8.6427	9.5425	10.982	12.338	14.041	17.240	21.337	26.039	30.813	33.924	36.781	40.289	42.796
23	9.2604	10.196	11.689	13.091	14.848	18.137	22.337	27.141	32.007	35.172	38.076	41.638	44.181
24	9.8862	10.856	12.401	13.848	15.659	19.037	23.337	28.241	33.196	36.415	39.364	42.980	45.559
25	10.520	11.524	13.120	14.611	16.473	19.939	24.337	29.339	34.382	37.652	40.646	44.314	46.928
26	11.160	12.198	13.844	15.379	17.292	20.843	25.336	30.435	35.563	38.885	41.923	45.642	48.290
27	11.808	12.879	14.573	16.151	18.114	21.749	26.336	31.528	36.741	40.113	43.195	46.963	49.645
28	12.461	13.565	15.308	16.928	18.939	22.657	27.336	32.620	37.916	41.337	44.461	48.278	50.993
29	13.121	14.256	16.047	17.708	19.768	23.567	28.336	33.711	39.087	42.557	45.722	49.588	52.336
30	13.787	14.953	16.791	18.493	20.599	24.478	29.336	34.800	40.256	43.773	46.979	50.892	53.672

1.2 T-Test Probability Table

1-tail	0.50	0.25	0.20	0.15	0.10	0.05	0.03	0.01	0.005	0.001	0.0005
2-tail	1.00	0.50	0.40	0.30	0.20	0.10	0.05	0.02	0.010	0.002	0.001
1	0.000	1.000	1.376	1.963	3.078	6.314	12.710	31.82	63.66	318.31	636.62
2	0.000	0.816	1.061	1.386	1.886	2.920	4.303	6.965	9.925	22.327	31.599
3	0.000	0.765	0.978	1.250	1.638	2.353	3.182	4.541	5.841	10.215	12.924
4	0.000	0.741	0.941	1.190	1.533	2.132	2.776	3.747	4.604	7.173	8.610
5	0.000	0.727	0.920	1.156	1.476	2.015	2.571	3.365	4.032	5.893	6.869
6	0.000	0.718	0.906	1.134	1.440	1.943	2.447	3.143	3.707	5.208	5.959
7	0.000	0.711	0.896	1.119	1.415	1.895	2.365	2.998	3.499	4.785	5.408
8	0.000	0.706	0.889	1.108	1.397	1.860	2.306	2.896	3.355	4.501	5.041
9	0.000	0.703	0.883	1.100	1.383	1.833	2.262	2.821	3.250	4.297	4.781
10	0.000	0.700	0.879	1.093	1.372	1.812	2.228	2.764	3.169	4.144	4.587
11	0.000	0.697	0.876	1.088	1.363	1.796	2.201	2.718	3.106	4.025	4.437
12	0.000	0.695	0.873	1.083	1.356	1.782	2.179	2.681	3.055	3.930	4.318
13	0.000	0.694	0.870	1.079	1.350	1.771	2.160	2.650	3.012	3.852	4.221
14	0.000	0.692	0.868	1.076	1.345	1.761	2.145	2.624	2.977	3.787	4.140
15	0.000	0.691	0.866	1.074	1.341	1.753	2.131	2.602	2.947	3.733	4.073
16	0.000	0.690	0.865	1.071	1.337	1.746	2.120	2.583	2.921	3.686	4.015
17	0.000	0.689	0.863	1.069	1.333	1.740	2.110	2.567	2.898	3.646	3.965
18	0.000	0.688	0.862	1.067	1.330	1.734	2.101	2.552	2.878	3.610	3.922
19	0.000	0.688	0.861	1.066	1.328	1.729	2.093	2.539	2.861	3.579	3.883
20	0.000	0.687	0.860	1.064	1.325	1.725	2.086	2.528	2.845	3.552	3.850
21	0.000	0.686	0.859	1.063	1.323	1.721	2.080	2.518	2.831	3.527	3.819
22	0.000	0.686	0.858	1.061	1.321	1.717	2.074	2.508	2.819	3.505	3.792
23	0.000	0.685	0.858	1.060	1.319	1.714	2.069	2.500	2.807	3.485	3.768
24	0.000	0.685	0.857	1.059	1.318	1.711	2.064	2.492	2.797	3.467	3.745
25	0.000	0.684	0.856	1.058	1.316	1.708	2.060	2.485	2.787	3.450	3.725
26	0.000	0.684	0.856	1.058	1.315	1.706	2.056	2.479	2.779	3.435	3.707
27	0.000	0.684	0.855	1.057	1.314	1.703	2.052	2.473	2.771	3.421	3.690
28	0.000	0.683	0.855	1.056	1.313	1.701	2.048	2.467	2.763	3.408	3.674
29	0.000	0.683	0.854	1.055	1.311	1.699	2.045	2.462	2.756	3.396	3.659
30	0.000	0.683	0.854	1.055	1.310	1.697	2.042	2.457	2.750	3.385	3.646
40	0.000	0.681	0.851	1.050	1.303	1.684	2.021	2.423	2.704	3.307	3.551
60	0.000	0.679	0.848	1.045	1.296	1.671	2.000	2.390	2.660	3.232	3.460
80	0.000	0.678	0.846	1.043	1.292	1.664	1.990	2.374	2.639	3.195	3.416
100	0.000	0.677	0.845	1.042	1.290	1.660	1.984	2.364	2.626	3.174	3.390
1000	0.000	0.675	0.842	1.037	1.282	1.646	1.962	2.330	2.581	3.098	3.300

1.3 Normal Distribution - Probability Table

z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
2.0	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817
2.1	0.9821	0.9826	0.9830	0.9834	0.9838	0.9842	0.9846	0.9850	0.9854	0.9857
2.2	0.9861	0.9864	0.9868	0.9871	0.9875	0.9878	0.9881	0.9884	0.9887	0.9890
2.3	0.9893	0.9896	0.9898	0.9901	0.9904	0.9906	0.9909	0.9911	0.9913	0.9916
2.4	0.9918	0.9920	0.9922	0.9925	0.9927	0.9929	0.9931	0.9932	0.9934	0.9936
2.5	0.9938	0.9940	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	0.9951	0.9952
2.6	0.9953	0.9955	0.9956	0.9957	0.9959	0.9960	0.9961	0.9962	0.9963	0.9964
2.7	0.9965	0.9966	0.9967	0.9968	0.9969	0.9970	0.9971	0.9972	0.9973	0.9974
2.8	0.9974	0.9975	0.9976	0.9977	0.9977	0.9978	0.9979	0.9979	0.9980	0.9981
2.9	0.9981	0.9982	0.9982	0.9983	0.9984	0.9984	0.9985	0.9985	0.9986	0.9986
3.0	0.9987	0.9987	0.9987	0.9988	0.9988	0.9989	0.9989	0.9989	0.9990	0.9990
3.1	0.9990	0.9991	0.9991	0.9991	0.9992	0.9992	0.9992	0.9992	0.9993	0.9993
3.2	0.9993	0.9993	0.9994	0.9994	0.9994	0.9994	0.9994	0.9995	0.9995	0.9995
3.3	0.9995	0.9995	0.9995	0.9996	0.9996	0.9996	0.9996	0.9996	0.9996	0.9997
3.4	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9998
3.5	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998
3.6	0.9998	0.9998	0.9999							